1. Zmodyfikowany kod z zajęć

```
function spline2D_comp()
  %spline2Duniform()
  spline2Dmatrix() % Draw matrix instead of uniform
return

function spline2Dmatrix()

% Exact parameters below
knot_vectorx = [...];
knot_vectory = [...];
coeffs = [...];

precision = 0.1

compute_nr_basis_functions = @(knot_vector,p) size(knot_vector, 2) - p - 1
mesh   = @(a,c) [a:precision:c]

px = compute_p(knot_vectorx)
tx = check_sanity(knot_vectorx,px)
nrx = compute_nr_basis_functions(knot_vectorx,px)
x_begin = knot_vectorx(1)
x_end = knot_vectorx(size(knot_vectorx,2))

x=mesh(x_begin,x_end);

py = compute_p(knot_vectory)
ty = check_sanity(knot_vectory,py)
nry = compute_nr_basis_functions(knot_vectory,py)
y_begin = knot_vectory(1)
y_end = knot_vectory(size(knot_vectory,2))

y=mesh(y_begin,y_end);

%X and Y coordinates of points over the 2D mesh
[X,Y]=meshgrid(x,y);

% Init grid as zeros
M = zeros(length(x), length(y));

for i=1:nrx
  vx=compute_spline(knot_vectorx,px,i,X);
  for j=1:nry
    vy=compute_spline(knot_vectory,py,j,Y);
    % Formula
    M = M + coeffs(i, j) * (vx .* vy);
  end
end

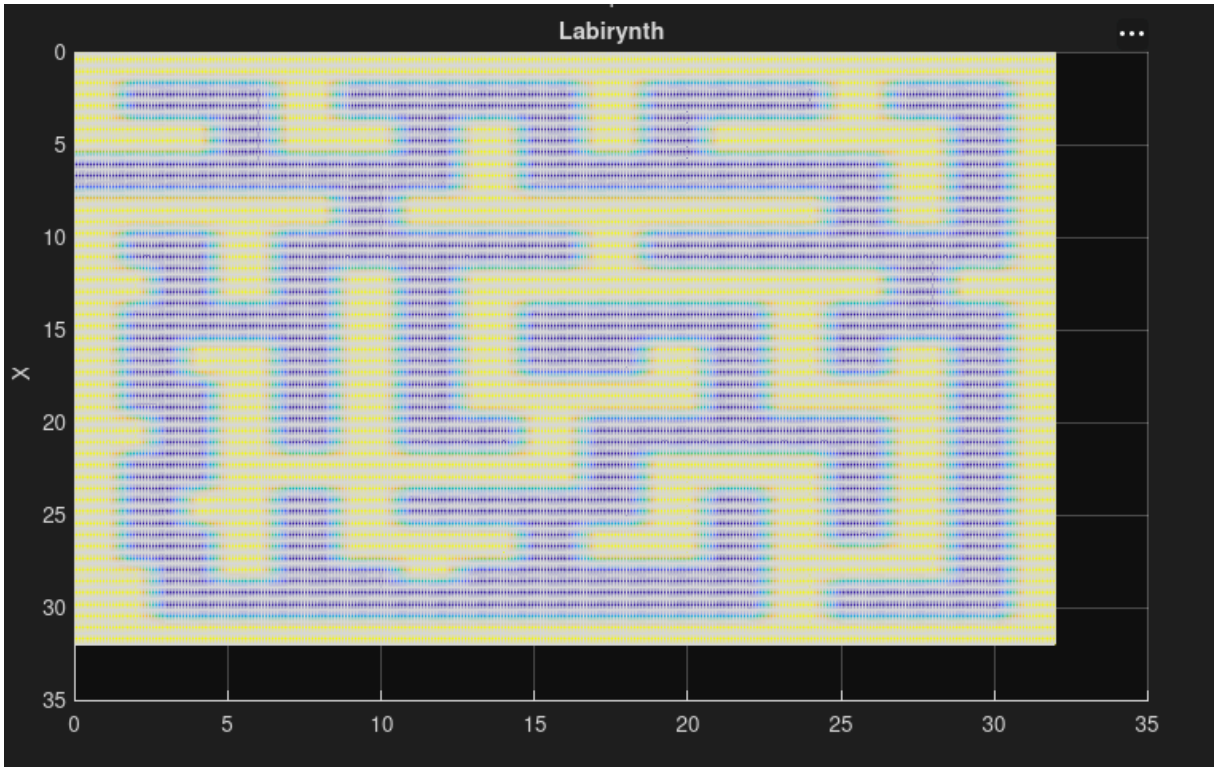
figure;
surf(X, Y, M);
xlabel('X');
ylabel('Y');
zlabel('Z');
title('Labirynth');
```

2. Wybrane parametry

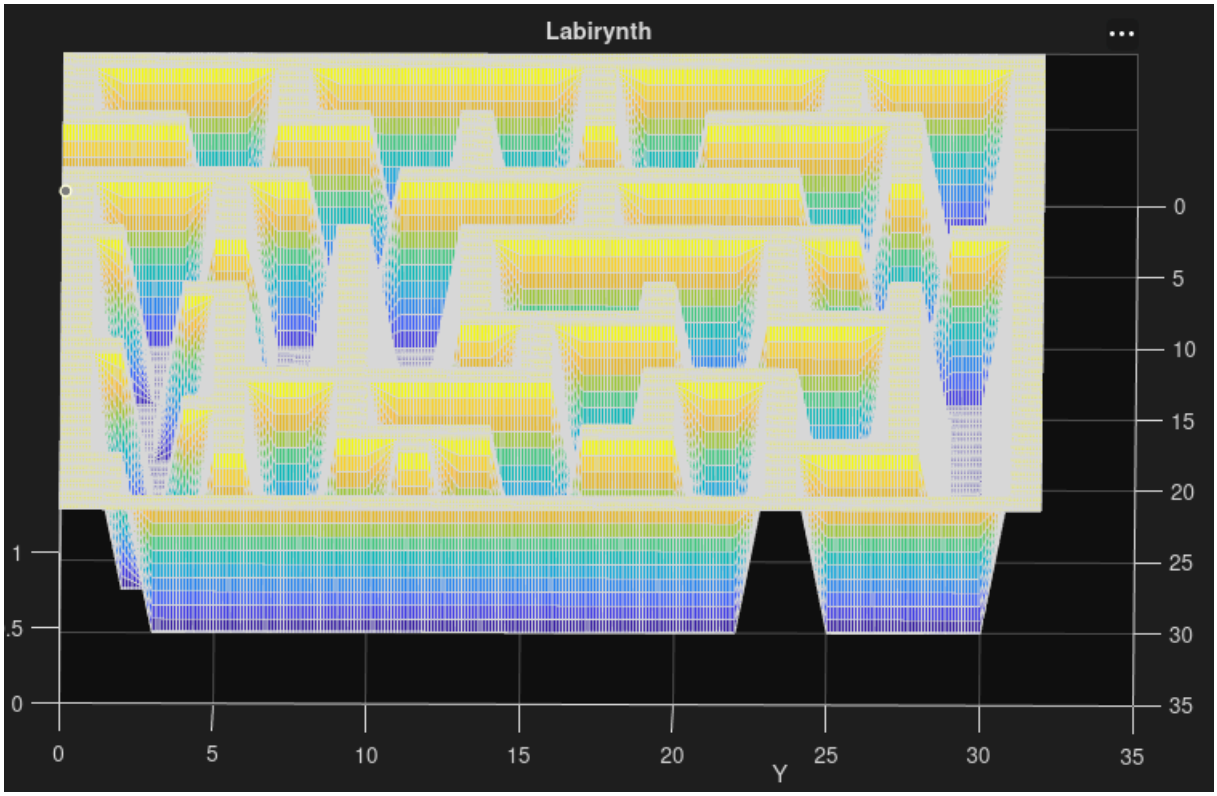
```
knot_vectorx = [0 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
26 27 28 29 30 31 32 32];
knot_vectory = [0 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
26 27 28 29 30 31 32 32];

coeffs = [
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1;
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1;
    1 1 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 1 1 0 0 0 0 1 1;
    1 1 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 1 1 0 0 0 0 1 1;
    1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 1 1 1 1 1 1 0 0 1 1;
    1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 1 1 1 1 1 1 0 0 1 1;
    0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 1 1;
    0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 1 1;
    1 1 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 1 0 0 1 1;
    1 1 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 1 0 0 1 1;
    1 1 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1;
    1 1 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1;
    1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 1 1 1;
    1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 1 1 1;
    1 1 0 0 0 0 0 0 0 1 1 0 0 1 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 1 1;
    1 1 0 0 0 0 0 0 0 1 1 0 0 1 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 1 1;
    1 1 0 0 1 1 1 0 0 1 1 0 0 1 1 0 0 0 0 1 1 0 0 1 1 0 0 1 1 0 0 1 1;
    1 1 0 0 1 1 1 0 0 1 1 0 0 1 1 0 0 0 0 1 1 0 0 1 1 0 0 1 1 0 0 1 1;
    1 1 0 0 0 1 1 0 0 1 1 0 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 1 0 0 1 1;
    1 1 0 0 0 1 1 0 0 1 1 0 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 1 0 0 1 1;
    1 1 1 0 0 1 1 0 0 1 1 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 1 1 0 0 1 1;
    1 1 1 0 0 1 1 0 0 1 1 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 1 1 0 0 1 1;
    1 1 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 1 0 0 1 1 0 0 1 1;
    1 1 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 1 0 0 1 1 0 0 1 1;
    1 1 0 0 1 1 1 0 0 1 1 0 0 0 0 0 0 0 0 1 1 0 0 1 1 0 0 1 1 0 0 1 1;
    1 1 0 0 1 1 1 0 0 1 1 0 0 0 0 0 0 0 0 1 1 0 0 1 1 0 0 1 1 0 0 1 1;
    1 1 0 0 0 1 1 0 0 1 1 1 1 1 1 1 0 0 1 1 1 1 0 0 1 1 0 0 1 1 0 0 1 1;
    1 1 0 0 0 1 1 0 0 1 1 1 1 1 1 1 0 0 1 1 1 1 0 0 1 1 1 1 1 1 0 0 1 1;
    1 1 1 0 0 1 1 0 0 0 0 1 1 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 0 0 1 1 1 1;
    1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 1 1 1;
    1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 1 1 1;
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1;
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1;
];
```

3. Uzyskany labirynt



Rysunek 1: Widok z góry



Rysunek 2: Widok z boku